

Satellite Realignment System

Operational Report — IMU Feedback + GPR Auto-tracking

Generated: 2026-05-05 10:27 UTC

1. System Overview

Parameter	Value
Stations Monitored	2
Online Stations	1
GPR Models Trained	0
IMU Feedback	Active (MPU-6050)
Architecture	ESP32 → FastAPI → Dashboard
Report Timestamp	2026-05-05 10:27 UTC

2. Station Status

Station	Status	Mode	AZ (°)	EL (°)	IMU AZ (°)	IMU EL (°)	Signal (dBm)	IMU Cal
Station A – Harare	ONLINE	Mode.MANUAL	65.0	111.5	0.0	0.0	N/A	No
Station B – Victoria Falls	OFFLINE	Mode.MANUAL	150.0	150.0	0.0	0.0	-20.0	No

3. Movement Verification Log

Time	Station	Result	IMU AZ Δ (°)	IMU EL Δ (°)	Target AZ	Target EL
2026-05-04T22:56	Station A – Harare	FAIL	7.91	0.00	0.0°	127.9°
2026-05-04T22:55	Station A – Harare	FAIL	35.94	0.00	0.0°	127.9°

4. Position vs Signal Log

Time	Station	AZ (°)	EL (°)	Signal (dBm)
2026-05-05T10:05	Station A – Harare	150.0	120.2	-11.0
2026-05-05T10:05	Station B – Victoria Falls	65.0	160.0	-14.0
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A

[illegible]

2026-05-05T08:40	Station A – Harare	65.0	160.0	N/A
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5. GPR Signal Model

Station	Trained	Samples	Trained at	Kernel (summary)
Station A – Harare	No	—	—	—
Station B – Victoria Falls	No	—	—	—

Top predicted angles (10° grid)

Station	Rank	AZ (°)	EL (°)	Predicted (dBm)	Confidence (%)
No data	—	—	—	—	—

6. Sweep Status

Station	State	Progress (%)	Positions	Started at	Finished at
Station A – Harare	SweepState.COMPLETE	0.0	0 / 0	—	—
Station B – Victoria Falls	SweepState.COMPLETE	100.0	25 / 25	2026-05-05T10:05	2026-05-05T10:10

7. Environmental Data

Station	Timestamp	Temp (°C)	Wind (km/h)	Rain	Humidity	Pressure
Station_2	2026-05-05T10:25:27.476580+00:00	30.2	11.2	0.0	40%	917
	2026-05-05T10:25:26.041412+00:00	24.7	10.6	0.0	39%	859
Station_2	2026-05-05T10:24:02.592254+00:00	30.2	11.2	0.0	40%	917
	2026-05-05T10:24:02.578918+00:00	24.7	10.6	0.0	39%	859
Station_2	2026-05-05T10:20:24.565642+00:00	30.2	11.2	0.0	40%	917
	2026-05-05T10:20:23.036499+00:00	24.7	10.6	0.0	39%	859
Station_2	2026-05-05T10:19:02.471305+00:00	30.2	11.2	0.0	40%	917
	2026-05-05T10:19:02.447287+00:00	24.7	10.6	0.0	39%	859
Station_2	2026-05-05T10:15:21.268186+00:00	30.2	11.2	0.0	40%	917
	2026-05-05T10:15:19.913871+00:00	24.7	10.6	0.0	39%	859
Station_2	2026-05-05T10:14:02.670151+00:00	30.0	10.7	0.0	41%	918
	2026-05-05T10:14:02.656226+00:00	24.5	10.4	0.0	40%	860
Station_2	2026-05-05T10:10:18.428238+00:00	30.0	10.7	0.0	41%	918
	2026-05-05T10:10:15.481841+00:00	24.5	10.4	0.0	40%	860
Station_2	2026-05-05T10:09:02.613526+00:00	30.0	10.7	0.0	41%	918
	2026-05-05T10:09:02.594681+00:00	24.5	10.4	0.0	40%	860
Station_2	2026-05-05T10:05:08.659852+00:00	30.0	10.7	0.0	41%	918
	2026-05-05T10:05:07.627271+00:00	24.5	10.4	0.0	40%	860
Station_2	2026-05-05T10:04:03.097006+00:00	30.0	10.7	0.0	41%	918

